Question

Electrochemistry is a fascinating branch of chemistry, where some substances undergo interesting reactions under the influence of electricity. For example, when electrolyzing acetic acid or sodium acetate solutions in different systems, the electrolysis products will vary depending on the most easily electron-losing part of acetic acid (sodium). Now there are the following statements:

1. When electrolyzing aqueous sodium acetate solution at normal temperature and pressure, 1.5 mol of gas will be produced at the anode for every 1 mol of electrons passed.

2. When electrolyzing aqueous sodium acetate solution at normal temperature and pressure, 0.25 mol of gas will be produced at the anode for every 1 mol of electrons passed.

3. When electrolyzing acetic acid in liquid HF solution at normal temperature and pressure, if the main product at the anode does not contain hydrogen and the mass fraction of C is 22.9%, then 0.167 mol of acetic acid is consumed for every 1 mol of electrons passed.

4. When electrolyzing acetic acid in liquid HF solution at normal temperature and pressure, if the oxidation product at the anode does not contain hydrogen and the mass fraction of C is 22.9%, then 0.125 mol of acetic acid is consumed for every 1 mol of electrons passed.

5. When electrolyzing acetic acid in liquid HF solution at normal temperature and pressure, if the oxidation product at the anode does not contain hydrogen and the mass fraction of C is 22.9%, then under this condition, acetic acid has two or more oxidation products, including $\mathrm{CO_2}$.

Which of the above statements are correct?

A. All statements are incorrect.

B. 1

C. 2

D. 3

E. 4

F. 5

G. 1,3

H. 1,4

I. 1,5

J. 2,3

K. 2,4

L. 2,5

M. 3,4

N. 3,5

O. 4, 5

P. 1,3,4

Q. 1,3,5

R. 1,4,5

S. 2,3,4

T. 2,3,5

U. 2,4,5

V. 1,3,4,5

W. 2,3,4,5

Answer

Correct Answer: G

Detailed Explanation

The reactions occurring during the electrolysis of aqueous sodium acetate solution are as follows:

Anode: $2CH_3COO^- \rightarrow C_2H_6 + 2CO_2 + 2e^-$

CHECKPOINT 1 PTS

$2CH_3COO^- \rightarrow C_2H_6 + 2CO_2 + 2e^-$

Cathode: $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$

CHECKPOINT 1 PTS

Cathode: $2H_2O + 2e^- \rightarrow H_2 + 2OH^-$

Therefore, for every 1 mol of electrons passed, the anode produces 0.5 mol of C_2H_6 and 1 mol of CO_2, totaling 1.5 mol of gas.

Statement 1 is correct, statement 2 is incorrect.

CHECKPOINT 1 PTS

For every 1 mol of electrons passed, the anode produces 0.5 mol of C_2H_6 and 1 mol of CO_2

During the electrolysis of acetic acid in liquid HF solution, it is known that the anodic oxidation product does not contain hydrogen, so the possible elements are only $C、O、F$. Let the chemical formula be Double subscripts: use braces to clarify. Based on the mass fraction, we can write:

$$\frac{12.01x}{12.01x+16y+19z} = 22.9\%$$

Increase x starting from 1. An integer solution appears starting at $x = 3$, and we calculate $x:y:z = 3:4:3$, i.e., $C_3O_4F_3$. At this point, a reasonable chemical formula cannot be written, so it is discarded.

An integer solution appears at $x = 4$, and we calculate $x:y:z = 4:3:6$, i.e., $C_4O_3F_6$. The only reasonable chemical formula that can be written is $(CF_3CO)_2O$.

Since the substrate is acetic acid, we temporarily do not consider complex products with a larger number of carbons. Consider the rationality of this product (trifluoroacetic anhydride):

In this system, acetic acid exists in molecular form and is not easily decarboxylated, but instead oxidizes the methyl side, replacing hydrogen with fluorine to become trifluoromethyl, "having different reactivity from aqueous sodium acetate solution", which is consistent with the conditions of the question. Therefore, this product is indeed the anodic oxidation product of acetic acid.

CHECKPOINT 2 PTS

The molecular formula of the anode product is $C_4O_3F_6$

CHECKPOINT 1 PTS

The chemical formula of the anode product is $(CF_3CO)_2O$

Therefore, the electrolysis equation can be written as:

$$2CH_3COOH + 6HF \rightarrow (CF_3CO)_2O + 6H_2 + H_2O$$

CHECKPOINT 1 PTS

$$2CH_3COOH + 6HF \rightarrow (CF_3CO)_2O + 6H_2 + H_2O$$

Or express H_2O as $H_3O^+ \cdot HF_2^-$

Then, for every 1 mol of electrons passed, $\frac{1}{6}$ mol of acetic acid is consumed. Statement 3 is correct, statement 4 is incorrect.

Acetic acid has only one oxidation product, and no CO_2 is produced. Statement 5 is incorrect.

Select G